Credit Card Fraud Detection

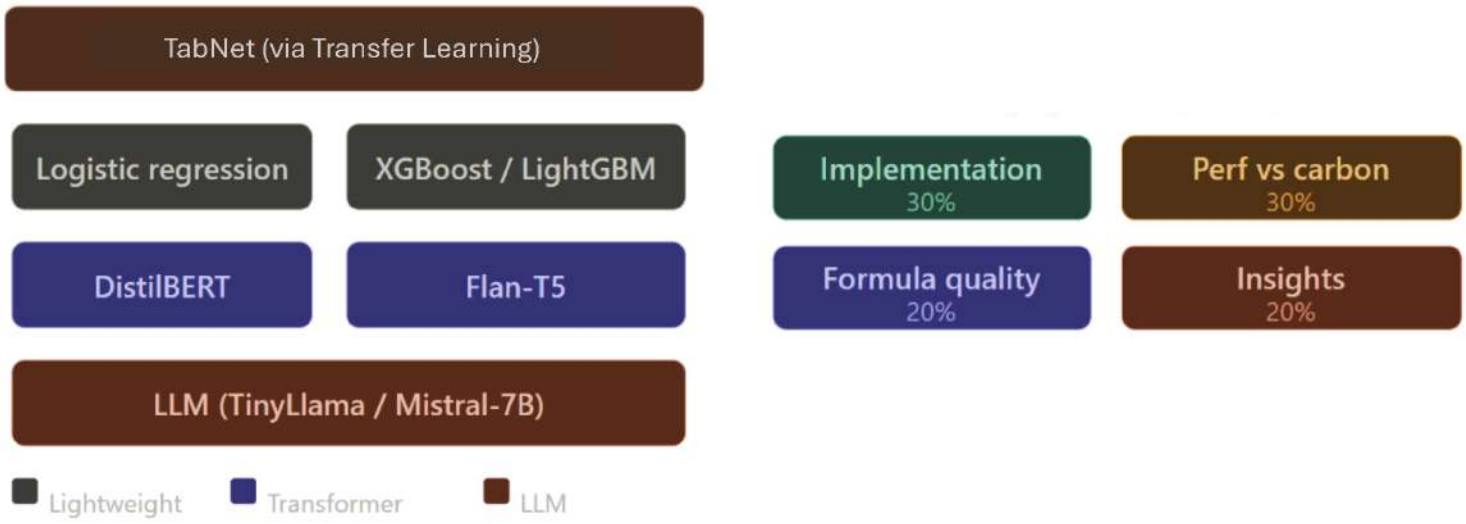
Team M3

Members: Seoyeon Kim, Amrit Rajeev, Mihir Bapat, Sakshi Khandelwal, Trisha Sharma

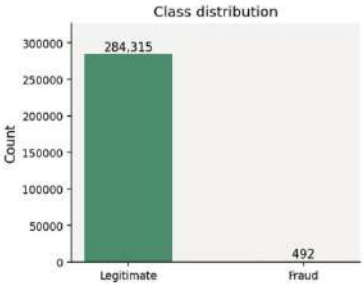

Step 1: Data prep
Kaggle fraud dataset
Step 2: Model selection
Tabular + transformer
Step 3: Train & evaluate
AUC, F1, precision/recall
Step 4: Carbon footprint
kWh × grid intensity
Step 5: Leaderboard
Performance ÷ carbon
Step 6: Presentation
Insights & recommendations

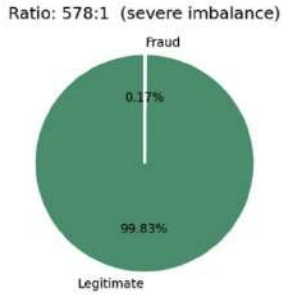
What we did

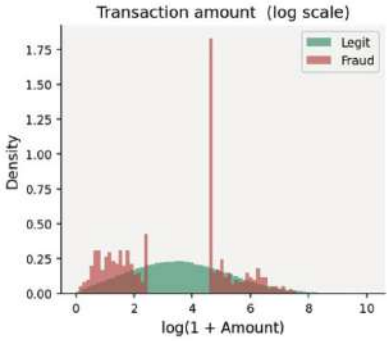
Model and evaluation

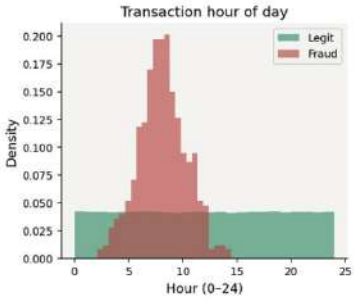


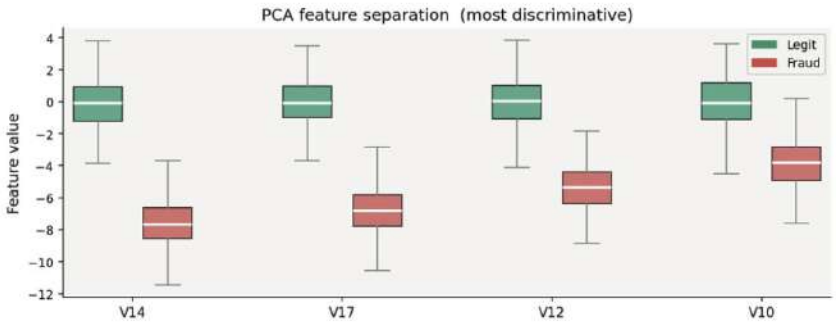
Dataset analysis & preprocessing guide

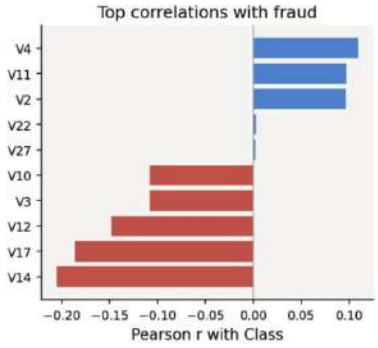












Results for different models

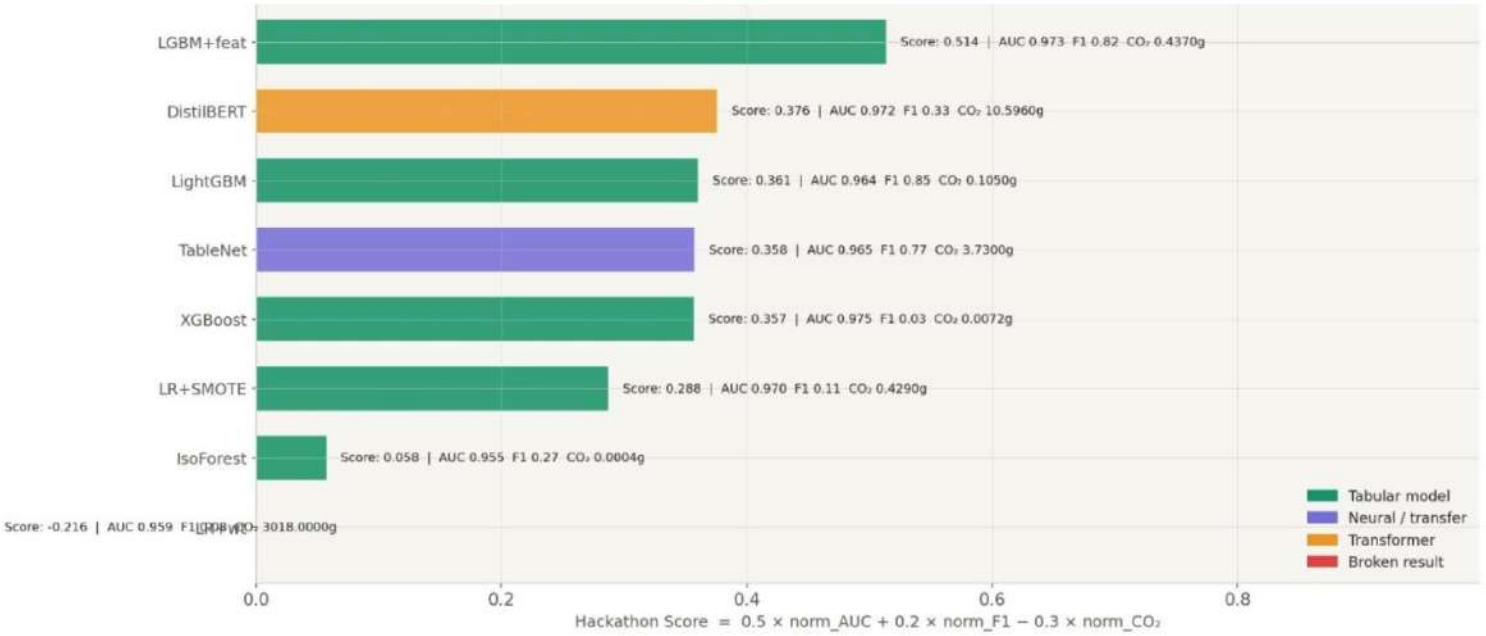
FULL LEADERBOARD — SORTED BY EFFICIENCY SCORE (F1 / NORMALISED CO_2)

Rank	Model	AUC	F1	Precision	Recall	Train CO_2 (g)	CO_2/1k preds (g)
1	**LightGBM**	0.964	0.850	0.833	0.867	0.105	0.00000106
2	LightGBM + features	0.973	0.820	0.803	0.836	437	0.000000447
3	TableNet (transfer)	0.965	0.766	0.721	0.816	3730	—
4	Random Forest	0.978	0.573	0.426	0.878	98.7	0.000092
5	RF + features	0.983	0.535	0.381	0.898	91.5	0.000094
6	DistilBERT (attempt 0)	0.972	0.330	0.200	0.900	10595	—
7	Isolation Forest	0.955	0.265	0.240	0.296	0.000356	0.000000001
8	LR + SMOTE	0.970	0.109	0.058	0.918	429	0.000000429
9	XGBoost	0.975	0.027	0.014	0.939	7.2	0.000018
10	LR + class weight	0.959	0.077	0.040	0.898	3,018,270	3018
11	DistilBERT (attempt 1)	0.706	0.000	0.000	0.000	11100	—

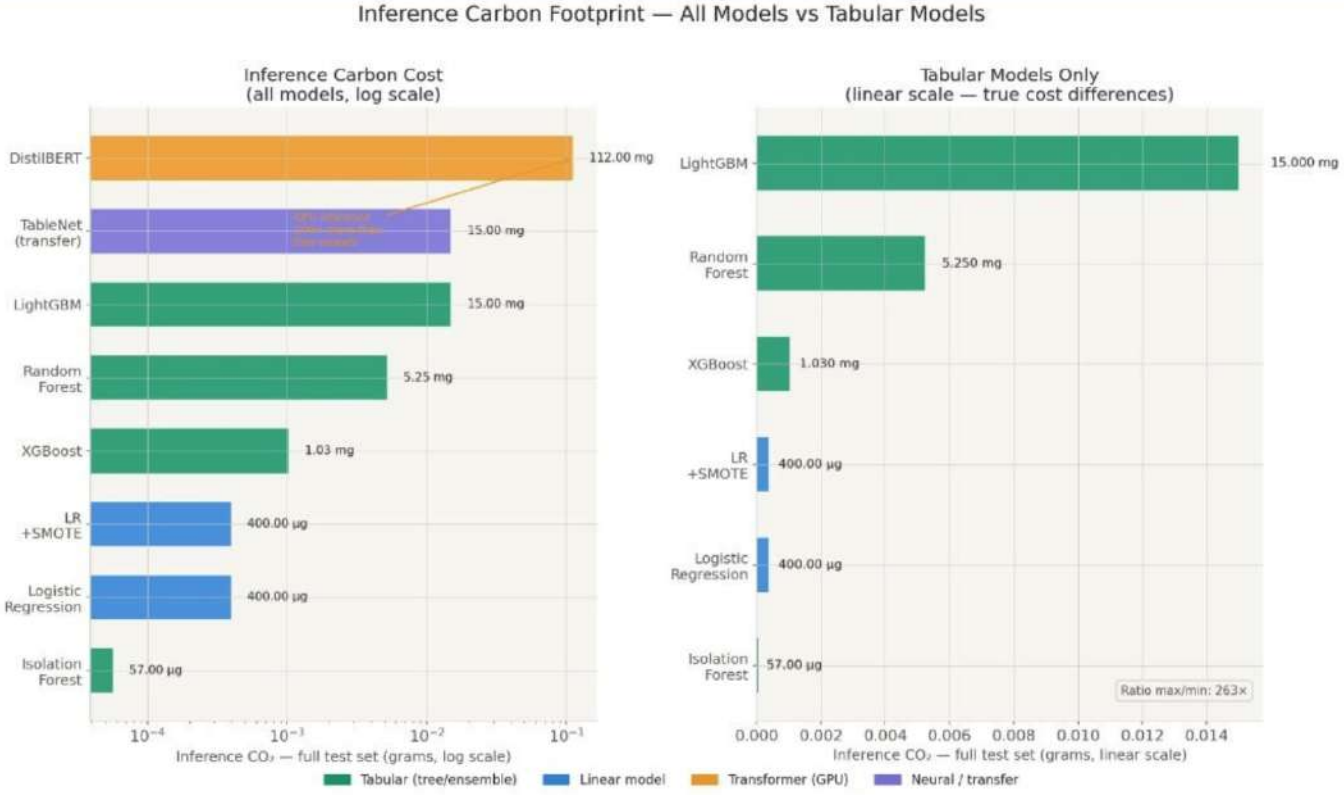
TOP-LINE METRICS — BEST MODEL PER CATEGORY

Best AUC	Best F1	Lowest CO_2 (train)	Best efficiency score
0.9825	0.850	0.000000356 g	LightGBM
RF + features	LightGBM	Isolation Forest	F1 ÷ CO_2 ratio

Leaderboard – Combined Efficiency Score



Inference Efficiency – F1 per Unit Inference Carbon



Which model gives the best performance per kg CO_2e?

LightGBM delivers the best real-world balance — AUC of 0.964, F1 of 0.850, and a training carbon cost of just 0.105g. Its efficiency score of 22.23 reflects strong performance at minimal environmental cost. It catches fraud reliably without the false alarm problem that plagues XGBoost (6,568 false alarms) or the carbon excess of heavier models. For a bank running millions of predictions daily, LightGBM is the only model that is both deployable and defensible.

Is a large LLM worth the carbon cost compared to smaller models?

No. DeBERTa consumed 10.6kg CO_2 in training — roughly 100,000 times LightGBM's footprint — yet scored a lower F1 of 0.330 and an efficiency score of just 0.24. LR+class_weight reached 3,018kg CO_2 with an F1 of only 0.077. On structured tabular fraud data, transformer and LLM-class models offer no performance advantage over gradient boosting. The carbon cost is not a tradeoff — it is pure waste.

What optimisations reduce carbon without hurting performance?

Three things made the biggest difference. First, choosing the right model family — tree-based models outperform neural approaches on tabular data at a fraction of the energy. Second, feature engineering — RF+features and LightGBM+feat show that domain knowledge (time-of-day and amount categories) can improve F1 without increasing model complexity. Third, threshold tuning — XGBoost's raw output was unusable (6,568 false alarms) but adjusting the decision threshold recovers precision at zero additional carbon cost. The lesson is that model selection and data quality are more carbon-efficient levers than model complexity.

Does deep learning outperform gradient boosting enough to justify its footprint?

No. TabNet Transfer Learning achieved F1 of 0.766 at 3.73g CO_2 — respectable, but LightGBM beats it on F1 (0.850) at 35× lower carbon cost (0.105g). DeBERTa scores worse on every metric while costing 100,000× more carbon. The gradient boosting family — particularly LightGBM — dominates this problem. Deep learning adds complexity, carbon, and longer training time in exchange for worse or equal results on tabular data.